

Q1

Answer the following questions by writing the queries in SQL and Relational Algebra.

- 1- Write an aggregate and group function operation to show the count of the employees and maximum salary from the EMPLOYEE table for each department by department number. (In SQL and Relational Algebra)

SQL:

Relational Algebra:

- 2- Retrieve all employees in department 5 whose salary > £30,000

SQL:

Relational Algebra:



Q2

Given the following EMPLOYEE file

EMPLOYEE (NAME , SSN , ADDRESS , JOB , SAL , ...)

In the above employee file, the record size is **R = 150 bytes** stored on the disk, block size is **B = 512 bytes** and total records in the file are **r = 30000** fixed length records

For an index on the SSN field in the employee file, assume the field size $V_{SSN} = 9$ bytes , assume the record pointer size **P_R = 7 bytes** .

You have to calculate the **Blocking Factor (Bfr)** for the **data file** as well as for the **index file** . Also calculate the **number of blocks** required to store data file and index file entries on the hard disk .

Also calculate and compare the **number of block accesses** for these files by using **binary search** and the **linear search** algorithms and **compare the results** of both the searching algorithms.

data file:

index file:

compare the results of both the searching algorithms:

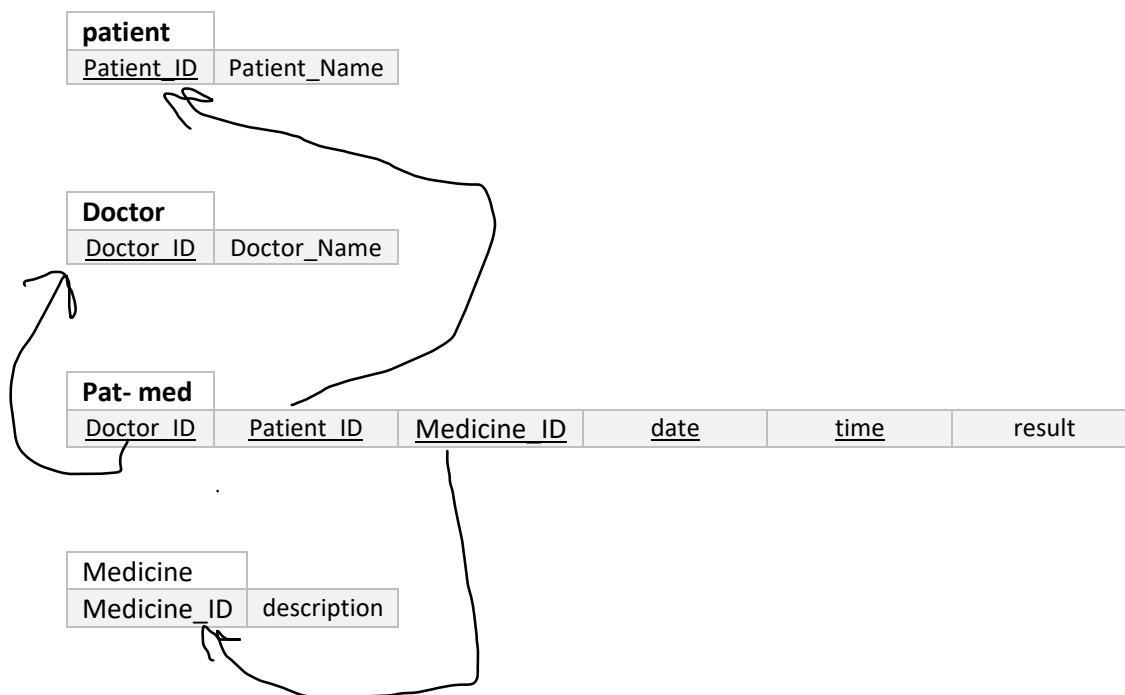


Q3

Relational Schema for the **hospital** Database is given in the following diagram. You have to map this **hospital** Relational Schema back into Entity Relationship Schema . In short you have to make the Entity Relationship Diagram (ERD) from the given Relational Schema . Properly draw the Entities , their attributes . Underline the Primary and Foreign Keys, make the Relationships between Entities using Referential Integrity Constraints and specify the type of the relationships between the Entities.

مهم تكتب ايش نوع العلاقة بينهم هل M-N or one to one or one to may تكون weak زي السؤال يلي جانا
فلازم تركز ع ذي و + ركز انه العلاقة يمكن

يمكن في اختلاف فالاسماء ولكن نفس الفكره



Q4

بعض الاختيارات ما أتذكر وش كانت الاجابه d فيها ولكن السؤال واضح فلو الاجابه مو من الخيارات المكتوبه دورها
فالسلايدات او فالنت

Relationship type where same entity type participates more than once in different place
called _____ relationship .

- a) One to Many b) Recursive c) one to one d) many to many

An attribute broken into component parts is called _____ attribute

- a) Multivalued b) Derived c) **Composite**

An attribute which can be calculated at run time is called _____

- a) Composite Attribute b) **Derived Attribute** c) Multivalued Attribute

A relationship in which three entities are participating is called _____

- a) Binary b) Ternary c) Quaternary

Degree of the relationship between four entities is

- A) Binary b) Unary c) Ternary

Which two of the following cannot be the appropriate entities ?

- a) System user b) Object c) System output



The following figure shows an ER schema for a database. Map this ER Schema into a Relational Schema, and specify all Primary and Foreign keys in the Relations. Make relationships between all the relations and specify the type of each relationship.

```

    erDiagram
        ATM ||--o{ Bank : Has
        ATM ||--o{ customer : Has
        Bank ||--o{ Branch : Has
        customer ||--o{ account : Has
        Transaction ||--o{ customer : Has

        ATM {
            string Cashlimit
            string ATM_ID PK
            string ATM_Location
        }
        Bank {
            string B_ID PK
            string B_Nmae
            string B_Add
        }
        Branch {
            string Br_Name
            string Br_ID PK
            string State
            string zip
            string Country
        }
        customer {
            string cust_ID PK
            string Country
            string Br_add
            string zip
            string State
            string Phone
        }
        Transaction {
            string Transaction_ID PK
            string Transaction_Type
        }
        account {
            string Acc_ID PK
            string Acc_Type
        }
  
```

Q6.1

كانت في 3 فقرات صح وخطا عن النورمالزيشن بس كانت مره سهله ومش متذكرها

Q6.2

الفقره الثانيه من السؤال السادس كانت انه عطانا جدول وطلب اننا نحوله لل Third normal form وكانت عن موظفين يشتغلون في بنك وعند كل موظف عقد واسم ورقم الفندق و مكان الفندق

SSN	Cont_id	Ename	Hotel_id	Hotel_loc
1234	B24	Ali	134	jeddah
1235	B25	Amar	135	abha
1236	B26	Ahmad	136	taif
1237	B24	Omar	134	jeddah

انتهت الأسئلة

للتواصل : <https://t.me/FCIT20Male>

موفقين

